

Electromagnetic Compatibility: MULTI-LAYER PCB DESIGNING



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In the first part we touched base on multi-layer PCB designing. In this part we discuss more about the same.

Multi-layer board and high-speed PCB design

A multi-layer board is a PCB having separate layers for power, ground and circuit tracks. It facilitates the use of entire copper layers for power and ground connections called power planes and ground planes, respectively. This decreases their inductance and impedance, allowing the use of high clock and signal frequencies and reducing ground noise. A rule of thumb is that, all other things being equal, a four-layer board will produce 15dB less radiation than a two-layer board.

New techniques employed in design and manufacturing of PCB circuits have, on one hand, made it possible to drastically reduce the size of the PCB. On the other hand, however, clock frequencies have also increased dramatically, resulting in very fast rise times (or short edge rates) of clocks. Such short edge rates mean that decreased PCB dimensions have started to have significant effects on signal voltages and currents.

High-speed circuits are those circuits that use clocks of fast (or small) rise times. But how does one define high speed? Well, the term high speed is relative, where one is essentially comparing two speeds. Take clocks, for example. One of these two

speeds is the rise time of the clocks, while the other is the speed of propagation of the electrons of the current that constitutes the clocks.

Now, the speed of movement of electrons in a PCB trace is less than the speed through air (which is the speed of light) since the field surrounding the electron has to travel through the PCB material that creates a drag. The equation given below gives the relation between speed of wave V_w along a PCB trace surrounded by PCB material to its speed C through a trace surrounded by air. ϵ_r is relative permittivity of the PCB material.

$$V_w = \frac{C}{\sqrt{\epsilon_r}}$$

For example, speed of an electromagnetic field in a trace surrounded by air is about 30.5cm (12-inch) per nanosecond (ns). But when the trace is laid on an epoxy PCB (with ϵ_r of 4), speed decreases to 15.2cm (6-inch) per nanosecond.

Now, consider a clock whose rise time is 1ns, and if dimensions of a PCB are greater than half the length traversed by this clock in 1ns (that is, greater than 7.6cm or 3-inch), track lengths will start to have significant effects on voltages. Let us see how.

Fig. 6 shows a typical clock with rise time of 1ns. In this time, current associated with this clock will travel 15.2cm (6-inch) along the track. Now, if the PCB dimension is less than 7.6cm (3-inch), reflected wave (wave will be invariably reflected if track changes direction or there is increase or decrease in track width) will arrive in less than 1ns (point X in Fig. 6), that is, at a time when the driver is still ramping the pulse up into the trace.

Because of driver inertia, reflected pulse will not have any effect on incident pulse. But if PCB dimensions (or track dis-

Fig. 6: Propagation delay

